



PADBen :

A Comprehensive Benchmark for Evaluating AI Text Detectors Against Paraphrase Attacks

⚠ Addressing the Critical Vulnerability

📋 4 Benchmark Tasks

🗄 Multi-Domain Dataset

📈 Standardized Evaluation

The Urgent Problem: Paraphrase Attack Vulnerability

⚠ Catastrophic Performance Collapse

Current AI text detectors show a dramatic decline in performance when facing paraphrase attacks, with accuracy dropping from **over 90%** to levels approaching **random chance**.

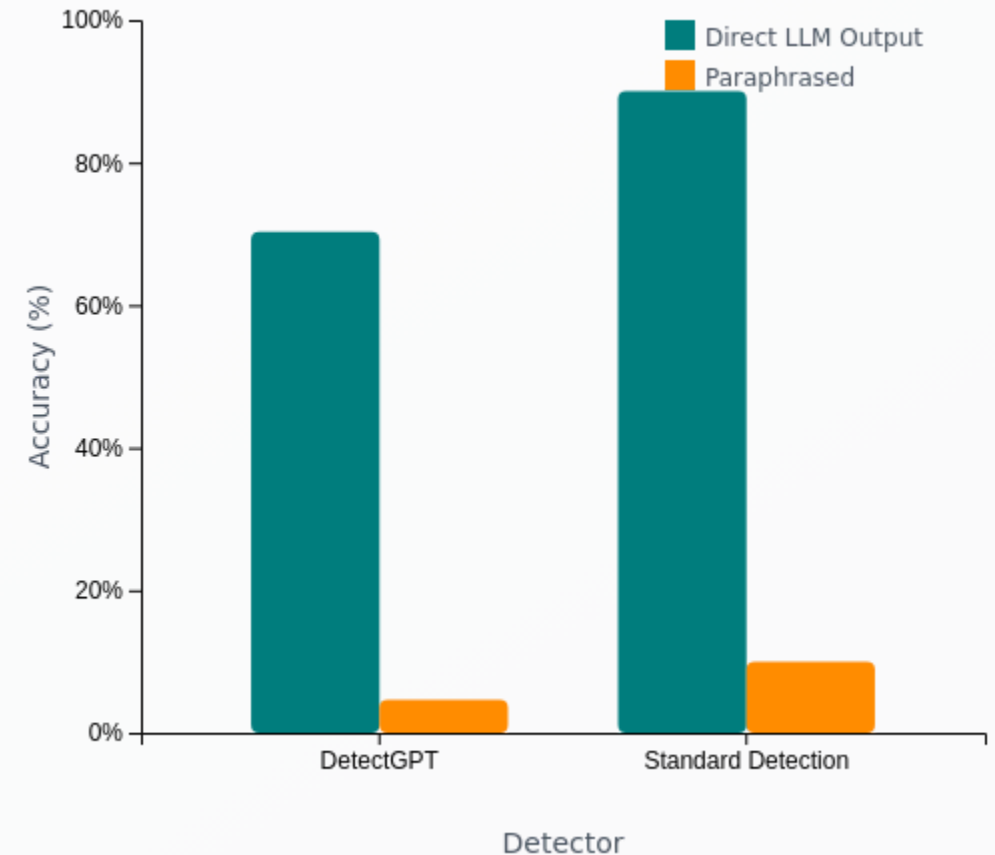
✓ Direct LLM Output

✗ Paraphrase Attacks

🛡 Security Implications

- 👤 Adversaries can easily "humanize" AI-generated content to bypass detection
- 🎓 Academic integrity threatened by undetected AI-generated content
- 🌐 Sophisticated misinformation campaigns can evade detection
- 🔒 Intellectual property protection compromised by paraphrased content

📊 Performance Degradation



Source: Krishna et al., 2023; Sadasivan et al., 2023

The Benchmark Gap

! The Critical Gap

Despite paraphrase attacks emerging as the primary evasion technique in 2024-2025, **no standardized benchmark** exists to evaluate detector robustness against them.

This gap leaves developers and users without reliable ways to assess or compare the resilience of different detectors in realistic threat scenarios.

🔍 Limitations of Existing Approaches

- ✔ **Binary Distinction:** Existing benchmarks (Mitchell et al., 2023; Bao et al., 2024) focus only on distinguishing between direct AI output and human-written text
- ✔ **Limited Scope:** More recent efforts like RAID (Dugan et al., 2024) expanded to include multiple adversarial attacks but lack focus on paraphrase nuances
- ✔ **Research Roadblock:** Without standardized evaluation, detector developers cannot systematically identify strengths and weaknesses across different paraphrasing strategies

🔗 Why This Gap Matters



Key Implications



Research

Stunted advancement of detection algorithms without rigorous testing



Security

Vulnerability to sophisticated evasion techniques remains unaddressed



Comparison

Impossible to objectively compare different detection approaches



Progression

Development of next-generation detectors hindered by lack of standards

PADBen: Overview & Contributions

“ Introducing **PADBen** (Paraphrase Attack Detection Benchmark), the first comprehensive benchmark specifically designed to evaluate the robustness of AI text detectors against a wide array of paraphrase attacks.



Comprehensive Benchmark Suite

Features four complementary tasks that simulate real-world detection scenarios, from identifying the paraphraser of a sentence pair to attributing the authorship of isolated text.



High-Quality, Diverse Dataset

A meticulously curated collection of multi-domain texts, including verified human paraphrases and iterative paraphrase chains, to test detector degradation under increasingly sophisticated attacks.



Systematic Baseline Evaluation

A thorough assessment of existing detectors—from traditional feature-based models to state-of-the-art systems—to systematically expose their vulnerabilities and identify architectural strengths.



Open and Accessible Resources

The public release of the complete benchmark dataset, standardized evaluation scripts, and a leaderboard to track community progress and foster reproducible research.



Real-world impact: The development of paraphrase-robust detectors is a timely and critical challenge. With the recent emergence and refinement of paraphrase attacks, the tools designed to ensure digital trust are being rendered ineffective.

Benchmark Design Principles

The PADBen benchmark is guided by three core principles designed to ensure its relevance, rigor, and real-world applicability. These principles work together to create a comprehensive evaluation framework for AI text detectors facing paraphrase attacks.



Authenticity

Ensures human-written text is genuinely human-authored, not assisted by AI.

- ✓ Human paraphrases verified as genuinely human-created
- ✓ Pre-LLM era datasets (MRPC from 2005) guarantee human authorship
- ✓ Manual verification of all human paraphrases



Diversity

Covers a wide range of domains, styles, and paraphrase strategies.

- ✓ Multi-source dataset from different domains
- ✓ Technical documentation, creative writing, social media, etc.
- ✓ Multiple paraphrase strategies and techniques



Realism

Reflects the actual evasion techniques employed against AI text detectors.

- ✓ Simulates real-world paraphrase attacks
- ✓ Iterative paraphrasing to test detection degradation
- ✓ Zero-shot and few-shot evaluation conditions



Why These Principles Matter

These principles ensure PADBen provides a standardized framework that effectively evaluates detector robustness against paraphrase attacks, closing the critical gap in existing evaluation methodologies.

Data Sources

Strategic selection of four complementary datasets to ensure comprehensive coverage of detection scenarios



MRPC

Microsoft Research Paraphrase Corpus (2005)

- ✓ **Rationale:** Pre-LLM era dataset ensuring human-written origins
- 📄 **Usage:** Extract sentence pairs labeled as semantically equivalent



HLPC

Human-LLM Paraphrase Collection (Lau & Zubiaga, 2024)

- ✓ **Rationale:** Verified human paraphrases of both human and LLM texts
- 📄 **Categories:** H-DOC, H-PP, LLM outputs and their paraphrases



OUTFOX

OUTFOX Essay Dataset (Verma et al., 2023)

- ✓ **Rationale:** Academic writing domain critical for integrity applications
- 📄 **Content:** Essays with human responses and commissioned human paraphrases



ODD

Original Diverse-Domain Dataset

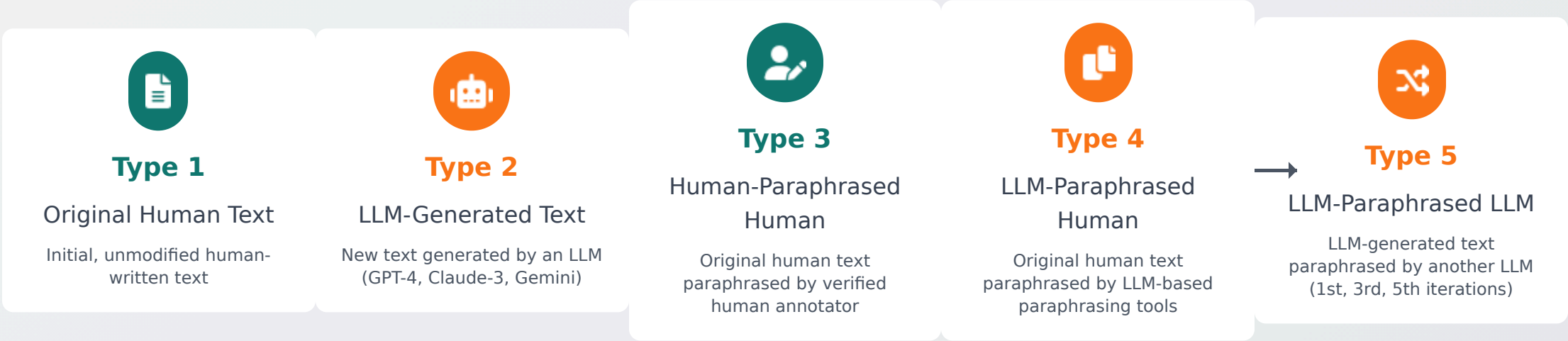
- ✓ **Rationale:** Fill gaps with underrepresented domains (technical, creative, social media)
- 📄 **Collection:** 2,500 human-written texts from 2023-2025 covering formal and informal registers

Key Benefits of Multi-Source Strategy

- ✓ Ensures human-written origins
- ✓ Covers multiple writing styles
- ✓ Includes iterative paraphrasing
- ✓ Balances formal and informal text

Paraphrase Generation Pipeline

For each base text selected from our sources, we construct a comprehensive set of five distinct text categories to simulate various real-world scenarios of human and AI text generation.



Simulation Purposes

- Human vs. AI Distinguishing**
Types 1-4: With and without human intervention
- Stylistic Artifact Detection**
Types 3-5: Identifying human vs. AI paraphrasing styles
- Paraphrase Chain Analysis**
Types 2-5: Tracking degradation across iterations
- Detector Resilience Testing**
Types 4-5: Evaluating robustness against sophisticated evasion

Task 1: Sentence Pair Paraphrase Detection

Task Definition

This task assesses a detector's ability to distinguish between human and LLM paraphrasing styles when the original text is available for comparison.

Input:

A pair of sentences, consisting of an original text and its corresponding paraphrase.

Output:

A binary classification: Human or LLM

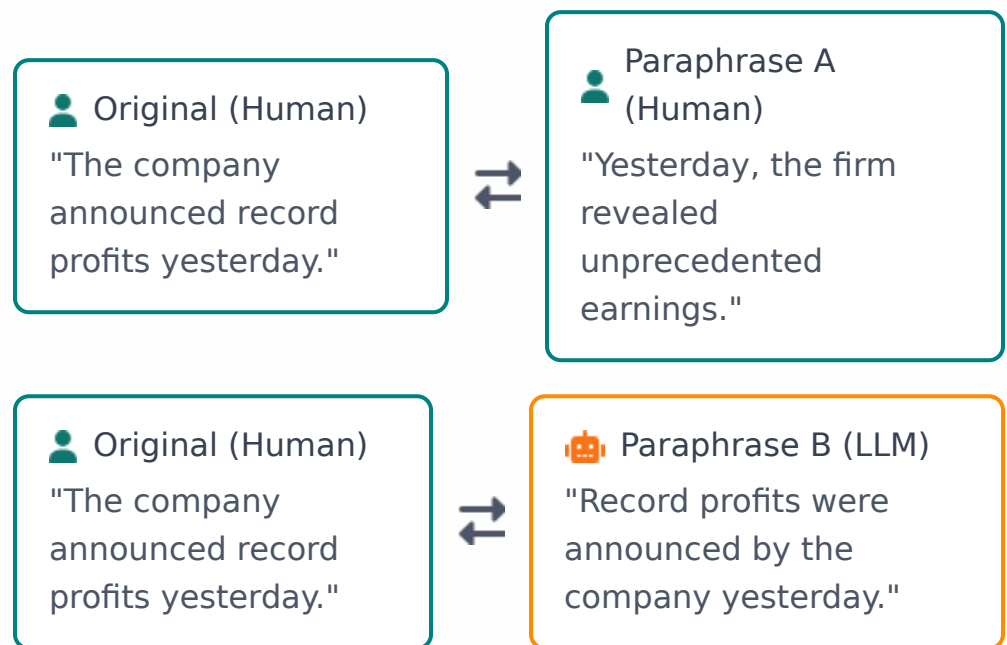
Key Insight

By providing the original sentence as context, this task focuses on identifying the subtle stylistic artifacts and patterns inherent in how LLMs rephrase text versus how humans do.



Leverages comparative analysis with the source text to detect subtle markers of AI vs human paraphrasing.

Example Illustration



Detection Goal

The goal is to correctly identify the author of Paraphrase A as **Human** and Paraphrase B as **LLM**.

-  Success requires detecting subtle stylistic differences between human and AI paraphrasing approaches.

Task 2: Single Sentence Paraphraser Detection

Task Definition

Input

A **single paraphrased sentence**, presented without its original source.

Output

A binary classification of the paraphraser's identity:
Human or **LLM**.

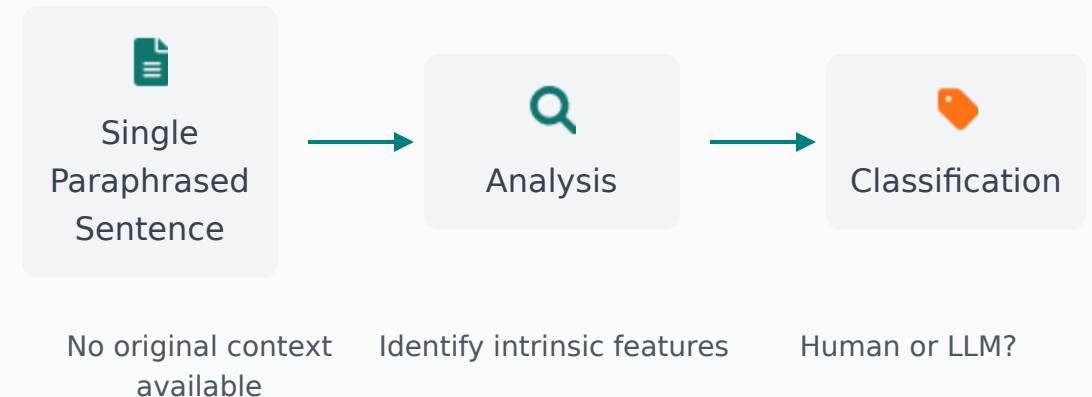
 **Key Insight:** Success in this task requires identifying inherent markers of LLM generation within a paraphrased text, as comparative analysis with the source is not possible.

Task Challenge

This task is fundamentally more challenging than Task 1 as it removes the comparative context, simulating a scenario where an investigator finds a suspicious text without its source.

 **Core Question:** Based on this text alone, was it paraphrased by a human or an LLM?

Task Flow



Example

"Record profits were announced by the company yesterday."

 Possible Human Paraphrase  Possible LLM Paraphrase

 The goal is to determine if the paraphrase was created by a human or an LLM based solely on the text itself.

Real-world scenario: Determining if a suspicious text found online was created by a human or automatically paraphrased by an AI.

Task 3: Single Sentence Authorship Attribution

Task Definition

Input:

Any single sentence, which could be an original human text, an original LLM text, a human paraphrase, or an LLM paraphrase.

Output:

A binary classification of the ultimate author: **Human-written** or **LLM-generated**.

” Example

"The company announced record profits yesterday."

Possible Origins:

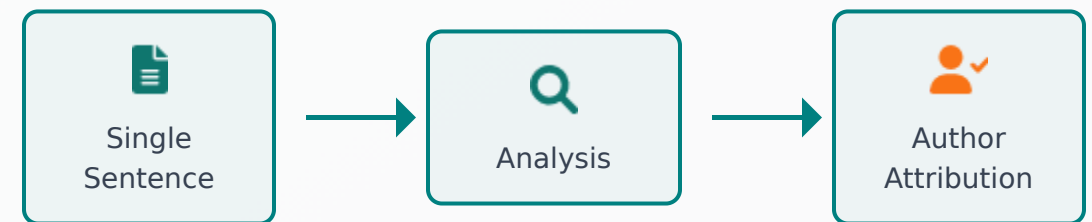
-  Original human text
-  Direct LLM output
-  Human paraphrase of any of the above
-  LLM paraphrase of any of the above

Task: Determine if the text is ultimately from a human or an AI.

Key Characteristics

- ✓ Most general and practical detection task
- ✓ No information about the text's origin or history
- ✓ Evaluates overall ability to attribute authorship in real-world scenarios
- ✓ Must be robust to both direct AI generation and various paraphrasing forms

Task Flow



 Most general form of detection, reflecting real-world scenarios

Task 4: Single Sentence AI Authorship Attribution

Task Definition

→ Input:

A sentence that is known to be generated by an LLM

→ Output:

Direct LLM output or LLM-paraphrased LLM output

Key Insight

This task measures a detector's sensitivity to the subtle changes introduced by iterative AI generation, specifically focusing on the "laundering" technique where AI-generated text is passed through multiple models to obscure its AI origin.

“ Critical for identifying attempts to obscure AI authorship by passing text through multiple models.

The "Laundering" Process



Example



GPT-4: "The company announced record profits yesterday."



Claude-3 paraphrase: "Record profits were announced by the company yesterday."



Task: Distinguish between direct GPT-4 output and GPT-4 output paraphrased by Claude-3

Task Comparison Table

Task	Input	Output	Key Question	Example
 Task 1 Sentence Pair Detection	Original sentence + Paraphrased sentence	Human LLM	Given the original, who paraphrased it?	Original: "The company announced record profits yesterday." A: "Yesterday, the firm revealed unprecedented earnings." B: "Record profits were announced by the company yesterday."
 Task 2 Single Sentence Detection	A single paraphrased sentence	Human LLM	Based on this text alone, was it paraphrased by a human or an LLM?	"Record profits were announced by the company yesterday." → <i>Determine if paraphrased by human or LLM</i>
 Task 3 Authorship Attribution	Any single sentence (original or paraphrased)	Human-written LLM-generated	Is this text ultimately from a human or an AI?	"The company announced record profits yesterday" → <i>Could be original human, LLM, human paraphrase, or LLM paraphrase</i>
 Task 4 AI Paraphrase Detection	A single LLM-generated sentence	Direct output Paraphrased output	Is this a direct AI output, or has it been laundered through another AI?	A sentence generated by GPT-4 → <i>Distinguish from direct GPT-4 vs GPT-4 paraphrased by Claude-3</i>

Baseline Detectors

PADBen evaluates a comprehensive array of detectors to establish robust performance baselines and identify architectural strengths and vulnerabilities.

⌚ Traditional Detectors



XGBoost

Feature-based model using linguistic features to distinguish between human and AI-generated text
Chen & Guestrin, 2016



RoBERTa & DeBERTa-v3

Fine-tuned large language models learning to classify text authorship based on deep contextual representations
Liu et al., 2019; He et al., 2021



Commercial Detectors

Widely used systems including OpenAI's detector and GPTZero, representing the state of commercially available tools
Solaiman et al., 2019; Tian, 2023

🚀 State-of-the-Art Detectors



FastDetectGPT

Zero-shot detector identifying AI text by analyzing probability curvature
Bao et al., 2024



Binoculars

Zero-shot method comparing cross-entropy of text under two different LMs to detect AI generation
Hans et al., 2024



RADAR

Detector specifically designed for adversarial robustness, trained to resist common evasion techniques
Hu et al., 2023



DetectLLM & GLTR

Zero-shot methods using text perturbations and statistical approaches to detect AI text
Su et al., 2023; Gehrmann et al., 2019

Evaluation Metrics

Primary Metrics



Accuracy

Overall percentage of correct classifications
 $(TP+TN)/(TP+TN+FP+FN)$



F1-Score

Harmonic mean of precision and recall, providing a balanced measure for imbalanced datasets



AUROC

Area Under the Receiver Operating Characteristic curve, assessing the model's ability to rank and distinguish between classes



TPR@1%FPR

True Positive Rate at a fixed 1% False Positive Rate, critical for high-stakes scenarios like academic integrity

Robustness Metrics



Paraphrase Degradation Rate (PDR)

Measures the percentage drop in performance when moving from evaluating direct LLM output to paraphrased text



Iterative Robustness Score (IRS)

Assesses performance consistency across multiple iterations of paraphrasing (1st, 3rd, 5th)



Cross-Domain Generalization (CDG)

Quantifies the variance in detector performance across different text domains (news, academic, social media)

Metrics Relationship

Primary Metrics

Robustness Metrics

The PADBen Score combines both primary performance metrics and robustness metrics to provide a comprehensive evaluation

$$\text{PADBen Score} = 0.3 \times \text{Avg_Acc} + 0.3 \times \text{Avg_AUROC} + 0.2 \times \text{Min_TPR@1\%FPR} + 0.2 \times (1 - \text{PDR})$$

Experimental Setup

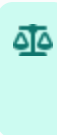


Training Protocol



Cross-Validation

5-fold cross-validation for robust performance estimation



Stratified Splits

Maintains original class balance in each fold



Early Stopping

Based on validation AUROC to prevent overfitting



Hyperparameter Tuning

Grid search over key parameters



Testing Conditions



Zero-shot

Pre-trained detectors evaluated directly on PADBen tasks without fine-tuning



Few-shot

Fine-tuning with limited examples (10, 50, 100) per class



Full Supervision

Models fully trained on task-specific data to assess maximum potential performance



Evaluation Flow



Data Processing



Model Training



Evaluation Metrics



Accuracy



AUROC



TPR@1%FPR



PDR



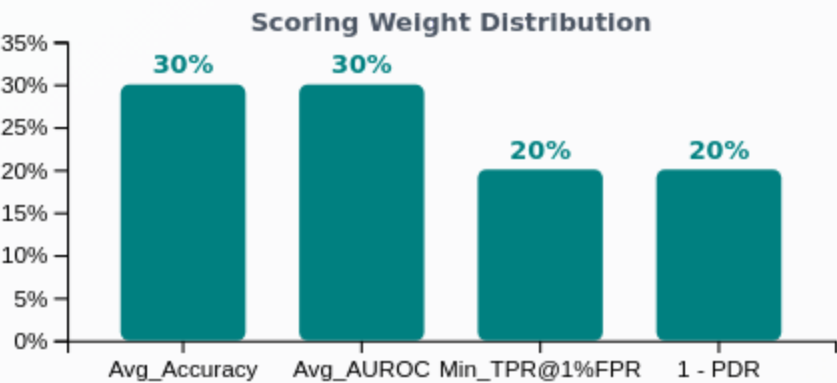
All experiments are conducted under controlled conditions to ensure reproducibility and fair comparison.

PADBen Score & Leaderboard

PADBen Score

A unified scoring system that summarizes overall performance, weighing both accuracy and robustness:

$$\text{PADBen Score} = 0.3 \times \text{Avg_Accuracy} + 0.3 \times \text{Avg_AUROC} + 0.2 \times \text{Min_TPR@1\%FPR} + 0.2 \times (1 - \text{PDR})$$



Why This Scoring Approach?

-  Balances average performance with worst-case robustness
-  Emphasizes resistance to paraphrase attacks
-  Focuses on security-critical applications

Leaderboard Categories

**Overall PADBen Score**

Ranks detectors based on combined score across all tasks

**Task-Specific Leaders**

Highlights top-performing detector for each of the four tasks

**Robustness Champion**

Recognizes best resistance to iterative paraphrase degradation

**Efficiency Leader**

Best trade-off between accuracy and computational requirements

Community Engagement

-  Public website (padben-benchmark.org) featuring:
-  Interactive leaderboard
 -  Dataset download portal
 -  Standardized submission system

Addressing Reviewer Concerns

In response to valuable feedback from reviewers, we've addressed key concerns to strengthen the benchmark's contribution.



Unclear Task Distinctions

Task definitions have been rewritten for clarity with explicit examples and explanations of key insights.

- ✓ Added comparison table to clearly delineate input, output, and objective of each task



Missing Baselines & Comparisons

Evaluation protocol has been expanded to include a comprehensive range of state-of-the-art detectors.

- ✓ Added FastDetectGPT, Binoculars, RADAR, DetectLLM, and GLTR to the evaluation



Incomplete Evaluation

Evaluation metrics now include critical security metrics and proper experimental design.

- ✓ Added TPR@1%FPR metric for security-sensitive applications
- ✓ Used stratified splits to handle dataset imbalances



Lack of Methodology & Limited Technical Contribution

Reframed the paper to explicitly position its contribution as a benchmark infrastructure.

- ✓ Emphasized the benchmark's role as an infrastructure contribution
- ✓ Added reproducibility resources and detailed experimental parameters

+ Additional Improvements

📄 Redesigned figures with larger fonts and clear labels 🌐 Public website with interactive leaderboard 🔄 Regular benchmark updates with new paraphrase models

Future Work & Community Engagement

PADBen's development represents the first comprehensive benchmark for paraphrase attack detection. Our roadmap includes regular updates and community engagement to ensure the benchmark's long-term relevance and impact.



Benchmark Evolution

- Quarterly additions of new paraphrase models
- Annual expansion with new, underrepresented domains
- Version control for fair comparison

Next update: Q4 2025



Community Engagement

- Public website: padben-benchmark.org
- Interactive leaderboard and dataset portal
- Workshops at major NLP conferences
- Industry partnerships for benchmark adoption



Reproducibility

- Public release of benchmark data
 - Standardized evaluation scripts
 - Baseline results and parameters
 - Detailed experimental parameters and seeds
- All resources available under open licenses

Join the PADBen Community

By participating in the PADBen community, researchers and practitioners can contribute to and benefit from a growing ecosystem dedicated to advancing AI text detection technology. Together, we can develop more reliable systems for ensuring academic integrity, combating misinformation, and protecting intellectual property in an era of sophisticated AI-driven text manipulation.

